

Monica Muthukumaran

Software Engineer | Backend & Full-Stack | Java, Spring Boot, Kafka, MongoDB

Chennai, India | +91 73974 47519 | monicamuthukumaran7@gmail.com | linkedin.com/in/monica-muthukumaran

GitHub: github.com/monica-muthukumaran | Portfolio: portfolio.monicamuthukumaran7.workers.dev

SUMMARY

Software Engineer with 5 years of experience building backend and full-stack systems. At Citi, designed, built and solely own two production Java Spring Boot microservices connected by Kafka that classify ~140K wealth-product records end to end in under 10 minutes. Previously spent 3+ years at Infosys on performance and reliability for Angular and Electron applications. Builds AI products on a deterministic core: ProofAegis was a Top 100 Build at Google Patchamomma 2026. Open to backend, platform and AI engineering roles.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL

Backend: Spring Boot, Spring Data JPA, Spring Kafka, Microservices, REST APIs, FastAPI, Flask

Data & Messaging: Apache Kafka, MongoDB (aggregation pipelines), PostgreSQL, MySQL, Redis, Neo4j, BigQuery, Firestore

Cloud & DevOps: Docker, Kubernetes, OpenShift, Harness CI/CD, Google Cloud Run, Git, AppDynamics, Datadog

Frontend & Testing: Angular, Electron, HTML, CSS, JUnit, Karma/Jasmine, Lighthouse

AI & Platforms: Gemini, Google ADK (multi-agent), Palantir Foundry

Concepts: Data structures and algorithms (DSA), system design, event-driven architecture, idempotency, caching, sagas

PROFESSIONAL EXPERIENCE

Citi — Application Development Programmer Analyst 2

Apr 2025 – Present

Capital Markets, Wealth Product Taxonomy (backend and full-stack)

Chennai, India

- Designed a hierarchical, rule-driven taxonomy engine for Wealth Product Taxonomy (WPT) classification; prototyped it in Python/FastAPI to settle the data model, then re-architected it solo into two production Java Spring Boot microservices (classification and enrichment).
- Connected the services through Kafka (Spring Kafka, consumer groups, dead-letter topics) so enrichment back-pressure appears as consumer lag rather than failed classifications; the pipeline stamps taxonomy on ~140K product and asset records end to end in under 10 minutes.
- Own both services from design to production support: hierarchy resolution over MongoDB unit-tested against fixture trees, AppDynamics instrumentation and health endpoints, and OpenShift deployments through Harness CI/CD.
- Built a shared MongoDB utility library (Centralized Mongo Pipeline Runner) that keeps aggregation pipelines in one repository and runs them from a single place instead of inside each service.
- Own the Security Onboarding application on Palantir Foundry end to end (development, deployment, production support) and build several other Foundry applications.
- Contributed to a Git Diff Tracker that surfaces configuration and code changes between releases, and to shared Angular component libraries reused across applications.

Infosys — Digital Specialist Engineer

Dec 2021 – Mar 2025

Angular and Electron applications (web and Windows desktop)

Chennai, India

- Delivered secure, production-grade Angular and Electron applications, owning technical troubleshooting and performance optimisation.
- Reduced application downtime by 26% with a service-worker-based upgrade strategy for Angular applications.
- Improved application performance by 31% after building performance observability with Datadog, Lighthouse and the Performance API to locate bottlenecks.
- Packaged Electron apps as MSI/EXE installers for Windows with automated upgrade and rollback flows; built IndexedDB (Dexie) and file-based logging with crash reports so failures could be diagnosed on the user's machine.
- Led code reviews for coding standards and architectural compliance; contributed to quarterly PI planning (scope definition, user-story prioritisation).

Infosys — Digital Specialist Engineer Trainee

Sep 2021 – Dec 2021

Full-stack MERN training (MongoDB, Express, React, Node.js); built responsive React interfaces.

Mysuru, India

PROJECTS

ProofAegis — AI invoice-exception engine | Top 100 Build, Google Patchamomma 2026

[Portfolio](#)

Stack: Python, Flask, Google ADK, Gemini, Firestore, BigQuery, Google Cloud Run

- Built an accounts-payable exception engine whose five cross-case checks read the whole workspace to catch what a three-way match approves: duplicates under new invoice numbers, split instalments and rate creep inside tolerance.
- Orchestrated four Gemini agents (Google ADK) behind a Flask API, each with a deterministic parser fallback; matching, tolerance and financial-impact figures are computed in plain Python and never by a model.
- Added a trust ledger that checks every model-produced figure against the deterministic result and overwrites on mismatch; ran 320 labelled cases on a 321-invoice synthetic portfolio, where 9 of 13 cross-case-only exceptions had passed a full three-way match.

Distributed Social Platform — LinkedIn-style professional network

[Portfolio](#)

Stack: Java, Spring Boot, Microservices, Apache Kafka, Redis, Neo4j, REST APIs

- Built five Spring Boot services (user, post, feed, notification, auth) behind one API gateway, split by data ownership rather than by layer.
- Modelled connections as a Neo4j property graph so second-degree recommendations are a single traversal instead of chained relational self-joins.
- Post creation publishes a Kafka event and returns immediately; feed, notification and analytics consume it independently, with a Redis read-through cache on the feed read path.

Distributed Payment Gateway — Razorpay-style payments backend

[GitHub](#) | [Portfolio](#)

Stack: Java 25, Spring Boot 4.1, Spring Data JPA, PostgreSQL; designed around Kafka, Redis, Resilience4j

- Designed a 14-entity payment domain model in which idempotency is a database unique constraint, every amount is integer paise, the audit log is append-only and no CVV column exists anywhere in the schema.
- Designed the asynchronous path: Kafka between authorisation and ledger, settlement and notification consumers, saga compensation across services, and signed webhooks with bounded retries and a dead-letter table with replay.

CERTIFICATIONS & RECOGNITION

Certifications: Spring Boot 0 to 100, Cohort 5.0 – Coding Shuttle (2026) | Palantir Foundry Software Development Essentials | MongoDB SI Associate | Infosys Certified Front-End Web Developer | FastAPI – Udemy

Recognition: Top 100 Build, Google Patchamomma 2026 (ProofAegis) | Insta Awards at Infosys for client engagement and delivery | Outstanding ratings in annual performance reviews

EDUCATION

Bachelor of Engineering — St. Joseph's Institute of Technology, Chennai | CGPA 8.5/10

2017 – 2021